

Rareș Brujan

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EDUCATION

Graz University of Technology (TU Graz) <i>MSc Computer Science, Major in Visual Computing</i>	Mar 2026 - Present <i>Graz, Austria</i>
Graz University of Technology (TU Graz) <i>BSc Software Engineering and Management</i>	Oct 2022 - Feb 2026 <i>Graz, Austria</i>
Deutsches Goethe-Kolleg Bukarest <i>German-language education, Grades 1–12</i>	2010–2022 <i>Bucharest, Romania</i>

EXPERIENCE

Computer Vision Research Project - From Bark to Byte <i>TU Graz × AutoForst - Industry Research Collaboration</i>	2026 - Present <i>Graz, Austria</i>
- Built a configurable Python pipeline for processing real multi-view timber scanner images and metadata, with validation, automated tests, manual review stages, and dataset export. - Integrated pretrained TimberVision YOLO segmentation and OpenCV preprocessing, with integrity checks, visual QA, and GPU-accelerated inference where available. - Prepared and validated training-ready datasets for CNN and Vision Transformer classification experiments.	

PROJECTS

GuidanceAR - Bachelor's Thesis: Mobile AR & Computer Vision System <i>Unity, C#, Python, Flask, OpenCV, YOLO</i>	2024 - 2025
- Developed a client-server system linking a Unity/C# mobile AR client to a Python/Flask API for OpenCV/YOLO inference and structured detection results. - Generated 20,000+ synthetic training images with Unity Perception; built Python/OpenCV tooling for segmentation masks and YOLO polygon annotations.	
China Trip Journey - Offline-First Full-Stack Application <i>React, TypeScript, Supabase, PostgreSQL, Vite, Vitest</i>	2026
- Built an offline-first React/TypeScript application with service-worker caching, local persistence, queued synchronization, and a Supabase backend. - Built and tested a typed Excel ingestion pipeline; designed PostgreSQL tables with RLS enabled, and implemented passcode-checked Supabase RPC access.	

TECHNICAL SKILLS

Programming: Python, C, C++, C#, TypeScript, JavaScript, Kotlin, SQL
Software Engineering: Git, GitHub Actions, CI/CD, Linux, REST APIs, automated testing, Agile/Scrum
Backend / Data: Flask, PostgreSQL, Supabase, Firebase, Firestore
Development: Unity, React, Jetpack Compose, Vite, Vitest
Computer Vision / ML: PyTorch, OpenCV, YOLO, object detection, image segmentation, synthetic-data generation, GPU-accelerated training/inference

LANGUAGES

Romanian (Native), German (C1), English (C1)